

1. Define the following terms with examples:
(a) Phase (b) Component (c) Solubility limit (d) Microstructure
 2. Differentiate between **homogeneous** and **heterogeneous** systems.
 3. State and explain the **Gibbs Phase Rule**, and discuss its significance in phase equilibria.
 4. What information can be obtained from a **binary phase diagram**?
 5. Explain the difference between **equilibrium** and **non-equilibrium cooling** of alloys.
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6. Sketch and label a **binary isomorphous phase diagram**.
 - Indicate the **liquidus**, **solidus**, and **solvus** lines.
 - Explain how to determine the **composition of phases** using the tie-line rule.
 7. For a **Cu–Ni system**, explain what happens during solidification for an alloy containing 40 wt % Ni cooled slowly from the liquid phase.
 8. Define a **eutectic reaction** and state its general equation.
 - Use an **Al–Si** or **Pb–Sn** diagram to illustrate the reaction.
 9. Describe how the **cooling rate** affects microstructure in **non-equilibrium solidification**.
 10. Explain the relationship between **free energy**, **entropy**, and **phase stability**.
 11. Differentiate between metals, alloys, solid solutions, and intermediate phases.
 12. State the four Hume–Rothery rules governing solid solubility.
 13. Distinguish between ferrous and non-ferrous alloys, giving two examples of each.
 14. Write the chemical composition of pig iron, and describe its conversion into steel by
 15. (a) Basic Oxygen Furnace (BOF) (b) Electric Arc Furnace (EAF).
 16. Draw and label the iron–carbon phase diagram (up to 6 wt % C).
 17. Identify phases α (ferrite), γ (austenite), δ (ferrite), and Fe_3C (cementite).
 18. Mark important temperatures: 723 °C (eutectoid), 912 °C ($\alpha \rightarrow \gamma$), 1394 °C ($\gamma \rightarrow \delta$), 1539 °C (melting).
 19. Explain how **carbon content** affects **strength**, **hardness**, and **ductility** in plain carbon steels.
 20. Compare and contrast:
 21. **Low-carbon**, **medium-carbon**, and **high-carbon** steels (composition, microstructure, uses).

22. What are the major alloying elements in **low-alloy steels** and their effects on properties? (Cr, Mn, Mo, Ni, V).
23. Discuss how **heat treatment** (austenitizing → quenching → tempering) improves steel properties.
24. Differentiate between the following:
25. **Ferritic, Austenitic, and Martensitic** stainless steels
26. **Hot-work, Cold-work, and High-speed** tool steels
27. Describe the **microstructural constituents** (ferrite + cementite) in low-carbon steel at room temperature and their influence on mechanical properties.
28. What is **maraging steel** and why is it suitable for aerospace and missile components?